

Welcome to the Course

Anomaly Detection

- ✔ **Video:** Detecting Anomalies
5 min
- ✔ **Video:** Detecting Anomalies in MATLAB
6 min
- ✔ **Reading:** Installing Pre-Trained Models
10 min
- 📖 **Reading:** PCB Anomaly Detection with PatchCore and FCDD
1h

Anomaly Detection Assessment

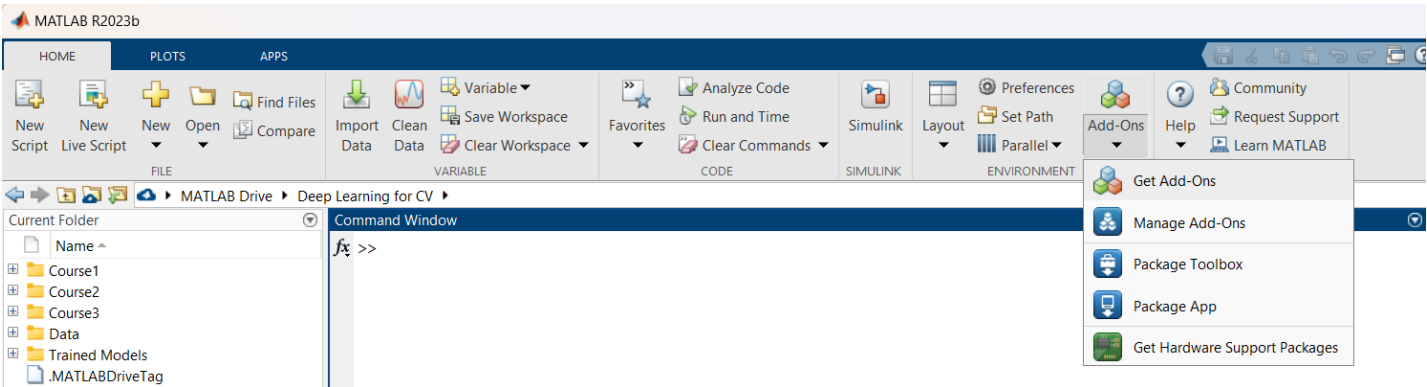
Installing Pre-Trained Models

To follow along and do the exercises in this course, you need to install three Add-On Packages to MATLAB. **If you completed Courses 1 and 2 of the specialization**, you already have these installed. If not, follow the instructions below.

Install the Computer Vision Toolbox Automated Visual Inspection Library

This add-on includes the anomaly detectors that you will use.

1. Go to the Add-On Explorer in the MATLAB Toolstrip and select "Get Add-Ons".



2. Search for "Computer Vision Visual Inspection Library" and select the add-on shown below. You will only see the green "Installed" banner if you've already installed the add-on.



Computer Vision Toolbox Automated Visual Inspection Library

by MathWorks Computer Vision Toolbox Team **STAFF**

Identify anomalies or defects in images to assist and improve quality assurance processes.

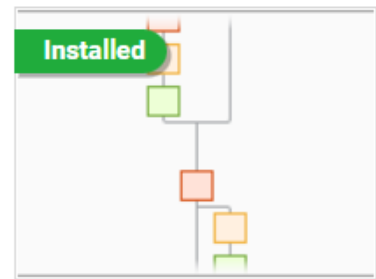
The **Computer Vision** Toolbox™ Automated **Visual Inspection** Library offers functions that enable you to train, evaluate, and deploy anomaly detection and object detection networks for use in **visual**

MathWorks Optional Feature

3. Follow the instructions to install the model.

Install the ResNet-18 pre-trained model

1. Go to the Add-On Explorer in the MATLAB Toolstrip and select "Get Add-Ons."
2. Search for "resnet-18 network" and select the add-on shown below.



Deep Learning Toolbox Model for ResNet-18 Network

by MathWorks Deep Learning Toolbox Team **STAFF**

Neural **Network** Toolbox Model for **ResNet-18 Network**

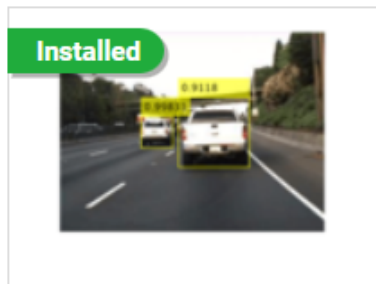
ResNet-18 is a pretrained model that has been trained on a subset of the ImageNet database. The model is trained on more than a million images, and can classify images into 1000 object categories

MathWorks Optional Feature

3. Follow the instructions to install the model.

Install the YOLOv4 pre-trained model

1. Go to the Add-On Explorer in the MATLAB Toolstrip and select "Get Add-Ons."
2. Search for "computer vision for yolo v4" and select the add-on shown below.



Computer Vision Toolbox Model for YOLO v4 Object Detection

by MathWorks Computer Vision Toolbox Team **STAFF**

Detect objects using a pretrained **YOLO v4** object detection network trained on COCO dataset.

objects from the COCO dataset. Refer to the **YOLO v4** documentation **for** more information.

MathWorks Optional Feature

3. Follow the instructions to install the model.

Go to next item ✓ **Completed**

